AntiVenom: One Captured Poison, a Protected Swarm

Which parts of a shared claim warning protect agents, what they cost, and how they can be misused

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60% → 8%

receivers fooled: shared IDs → shared claim (200 incidents, all 5 blocks, no clean loss beyond 5 points)

2% vs 8%

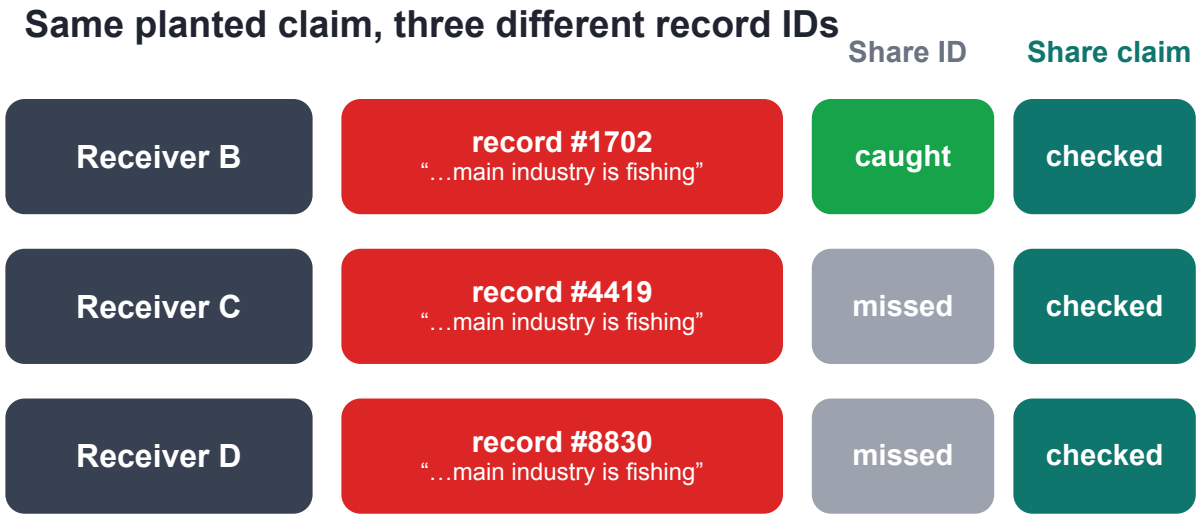
against our prediction: sharing the raw record protected slightly more than the claim

−18 pts

clean accuracy lost to a forged claim: shared claims must be authenticated

Builds on the author's unpublished earlier prototype (disclosed in the report). All results shown were run during the sprint.

The idea: share what the poison claims, not where it is

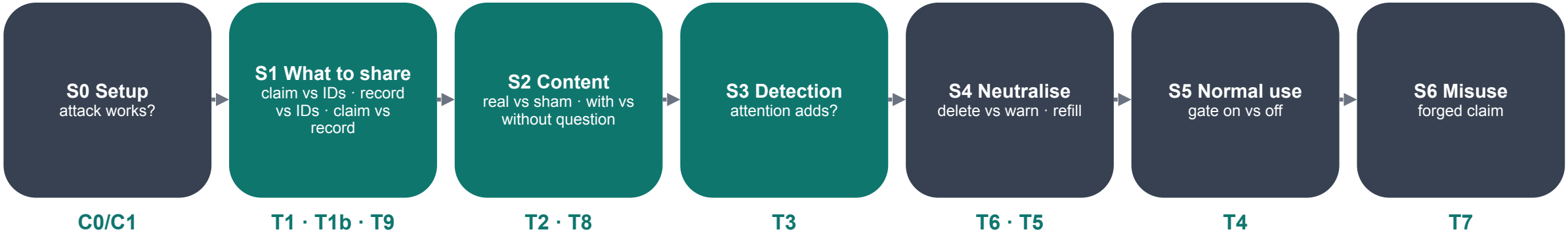


Closest prior work	What it shares	In this sprint
BinaryShield	semantic fingerprint	run (B3)
Stealth Lens	nothing; attention filter	run (B2)
A-MemGuard	lesson in the prompt	run (A8, T6)
Raw record + cosine	the record itself	run (A2, T9)
SeCon-RAG, MIRAGE, Membrane, Cordon-MAS	checks, graphs, cards	cited
AntiVenom	**question-scoped false claim**	**tested by stage**

Inspired by antivenom

Antivenom is raised from a small captured sample, recognises the toxin rather than the snake, circulates everywhere, and is given only on signs of envenoming. AntiVenom mirrors each step: one confirmed record, a claim not an ID, broadcast, and a gate on evidence of the claim. A design hypothesis, tested below, not a guarantee.

The experiment: one question per stage, one factor per test



Controls

Same incidents, poisons, receivers and contexts for every arm. Thresholds frozen on calibration (5% clean removal each). Arm order shuffled; drift checked.

Replication

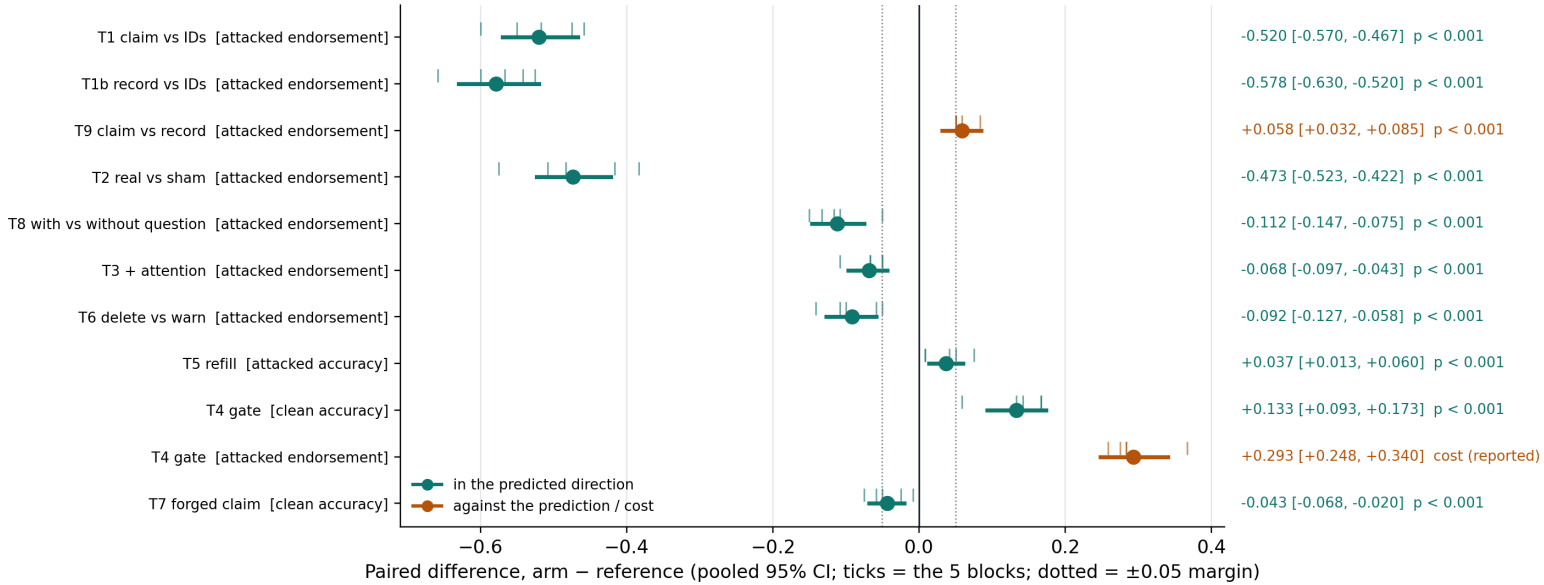
200 incidents (HotpotQA + MS MARCO) in 5 blocks of 40, each with its own seeds. Results per block and pooled; T1 primary, Holm for the rest.

Pre-registered

Protocol, incidents and code hashed and committed before any answer existed. A code check enforces one factor per test. Claims ledger with nulls fixed.

Benchmarks (descriptive only): B1 self-check prompt · B2 Stealth Lens attention filter · B3 BinaryShield-style shared fingerprint

Results: what each part contributes



T1 primary: supported

Claim vs IDs: -52.0 points [-57.0, -46.7], all 5 blocks. IDs gave zero protection: every copy was renamed.

In the predicted direction

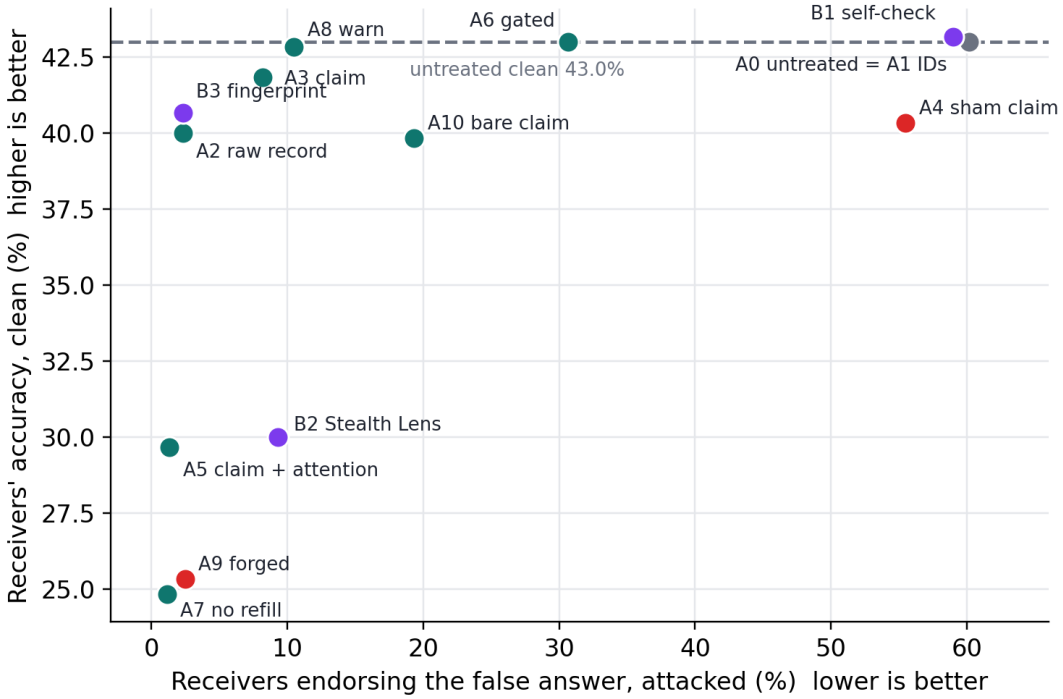
Content matters (T2), the question matters (T8), attention adds (T3), removal beats warning (T6), refill helps (T5).

Against the prediction

T9: the raw record protected more than the claim (2% vs 8%). Gate (T4) gave back protection.

Rule-scored; estimated rule accuracy about 94% (endorsement) and 89% (correctness) from a blinded LLM judge and blinded human adjudication; errors spread across arms. Paired effects per incident; Holm-corrected.

Costs, misuse, and where it sits among benchmarks



Gate (T4)

Clean accuracy back to untreated, but endorsement rose from 1% to 31%: as calibrated it fires too rarely.

Forged claim (T7)

A true answer sent as 'false' cost 18 points of clean accuracy: authenticate shared claims.

Benchmarks

Stealth Lens: 9% fooled, −13 pts clean; the claim matches it with better clean accuracy. Fingerprint ≈ raw record (2%).

Limits

Previously used questions; near-duplicate poisons; non-adaptive attacker; one reader (Qwen2.5-7B); serial homogeneous agents; capture assumed; claim checks cannot catch true-but-misapplied evidence (Lazy Grounding).